

Quantum Annealing for Active User Detection in Non-Orthogonal Multiple Access System

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① Introduction

② Problem definition

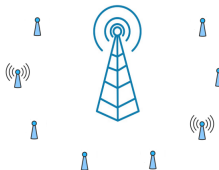
③ Quantum annealing approach

④ Analysis of the algorithm

⑤ Conclusion

What is AUD ?

In massive wireless networks, resource allocation prior to transmission is no longer adapted for 5G applications



Transmission protocol ¹:

- ① (Downlink) The AP indicates the beginning of the frame
- ② (Uplink) Active users send a pilot sequence
- ③ AP estimates the active users to start an allocation protocol

¹Jeannerot et al. Mitigating User Identification Errors in Resource Optimization for Grant-Free Random Access, 2023

Existing approaches

Leveraging Non-orthogonal multiple access recently appeared relevant in the context of AUD but it makes it a challenging task

→ AUD becomes an inference problem



Inferred variables are binary: combinatorial optimization

Quantum algorithms

Quantum algorithms gained significant interest for optimization tasks due to the potential speed up they might provide

- Grover's algorithm: from $\mathcal{O}(2^N)$ to $\mathcal{O}(2^{N/2})$
- Quantum annealing: quantum analogue of simulated thermal annealing
- Quantum heuristics: QAOA

Quantum annealing has been deeply studied, furthermore it is quite intuitive and it is nicely adapted to some class of combinatorial optimization problems

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Problem to solve

- N sequences of length $M < N$ distributed among the users
→ Non-orthogonality
- At the beginning of the frame, a random subset of the users transmit their sequence to the K -antenna access point:

$$\mathbf{Y} = \sum_{i=1}^N \alpha_i^{(0)} \mathbf{p}_i \mathbf{h}_i^T + \mathbf{W}$$

$\alpha_i^{(0)}$ indicates the state of the i -th user

- $\mathbf{h}_i$: Rayleigh fading with *unit large scale fading*

Problem to solve

- 1 How can we estimate the activity pattern $\alpha^{(0)}$ (ground truth) from the received signal ?

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- 1 How can we estimate the activity pattern $\alpha^{(0)}$ (ground truth) from the received signal ?
- 2 Does the choice of the sequences $\mathbf{p}_i$ has an impact on the difficulty of the problem ?

Covariance based AUD

The *sample covariance* of the activity pattern is a sufficient statistics for the activity pattern recovery²:

$$\hat{\Sigma}_Y = \frac{1}{K} \sum_{k=1}^K \mathbf{y}_k \mathbf{y}_k^H = \frac{1}{K} \mathbf{Y} \mathbf{Y}^H$$

It motivates the introduction of the *NNLS estimator*:

$$\alpha^{\text{NNLS}} = \arg \min_{\alpha \in \{0,1\}^N} \left\| \mathbf{P} \cdot \text{Diag}(\alpha) \cdot \mathbf{P}^H - \hat{\Sigma}_Y \right\|_F^2$$



Idea: match the true covariance to the sample one

²S. Haghighatshoar, P. Jung, and G. Caire, “Improved Scaling Law for Activity Detection in Massive MIMO”, 2018

Pilot scheme

The pilot matrix $\mathbf{P}$ impacts the structure of the function to minimize
→ Does it have an impact on the efficiency of quantum annealing ?

For the sake of simplicity, we propose the two random design schemes:

$$\mathbf{p}_i \sim \begin{cases} \mathcal{N}\left(0, \frac{1}{\sqrt{M}} I_M\right) & \text{(Gaussian)} \\ \mathcal{U}\left(S^{M-1}\right) & \text{(Unit sphere)} \end{cases}, \quad i = 1, \dots, N$$

Large number of antenna

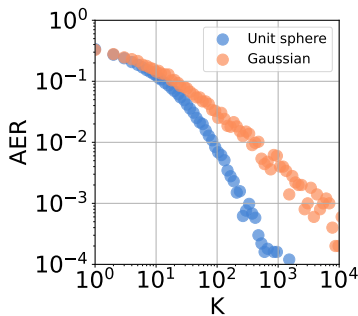


Figure 1: AER^{NNLS} against the number of antenna K at the AP for Gaussian and unit sphere schemes, at $\text{SNR} = \infty$



We adopt the regime $K \gg 1$

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Ising Hamiltonian

The NNLS computation is equivalent to finding the *ground state* of the Ising Hamiltonian:

$$\hat{H}_P(\sigma) = - \sum_{i < j} J_{ij} \hat{\sigma}_i^z \hat{\sigma}_j^z - \sum_i b_i \hat{\sigma}_i^z$$

- The quantum spins $\hat{\sigma}_i^z$ are directly related to the binary variables α_i
- The parameters J_{ij} and b_i depend on a problem instance (P, Y)

Principle of quantum annealing

One defines the quantum Hamiltonian:

$$\hat{H}(u(t)) = (1 - u(t))\hat{H}_P + u(t)\hat{H}_C$$

- $\hat{H}_P$ contains the problem to solve, $\hat{H}_C$ introduces quantum fluctuations between the different spin configurations
- The *control function* $u(t)$ starts at $u(0) = 1$ and decreases towards $u(T) = 0$ after a certain time T

Scheduling quantum annealing

The decay rate of the control function must obey a stringent condition, known as the *adiabatic theorem*.

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→ Evolution controlled by *the instantaneous gap* $\Delta_{(P,Y)}(u(t))^2$

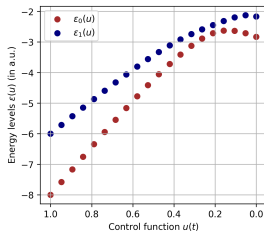
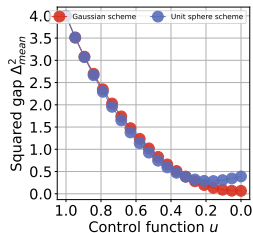


Figure 2: Energy levels during the quantum annealing process

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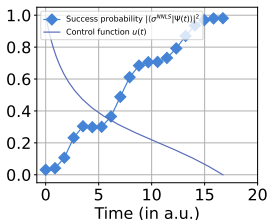
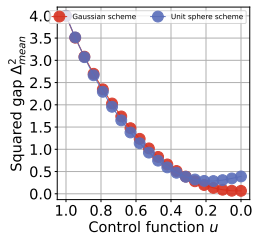
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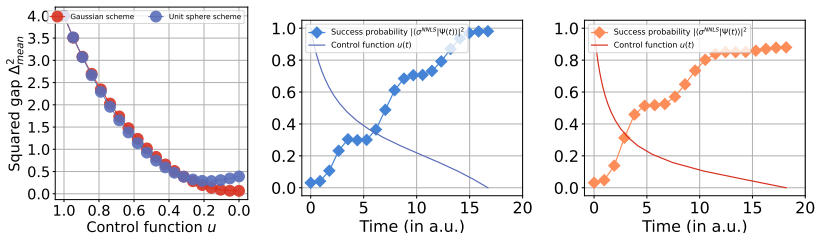


Figure 3: Averaged spectral gaps and example of QA process for each pilot scheme

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Quantum activity-error rate

The success probability is a vector-wise metric. Isn't it too detrimental ?

However, we face some challenges to build a component-wise metric:

- 1 Take into account probabilistic nature of quantum measurements
- 2 The outputs are ± 1

Quantum activity-error rate

The success probability is a vector-wise metric. Isn't it too detrimental ?



We introduce the *quantum activity-error rate*:

- Correlation of the output:

$$d = \frac{1}{N} \sum_{i=1}^N \sigma_i^{(0)} \langle \Psi(T) | \hat{\sigma}_i^z | \Psi(T) \rangle$$

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- qAER:

$$\text{qAER} = \mathbb{E}_{(\mathbf{P}, \mathbf{Y})} \left(\frac{1 - d}{2} \right)$$

Noise sensitivity

- We checked the behavior of the qAER metric against the SNR
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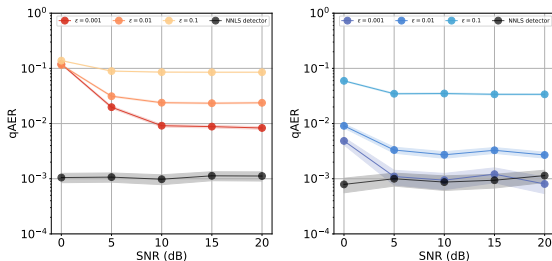


Figure 4: Quantum activity-error rate for Gaussian scheme (left) and unit sphere scheme (right) with precision levels $\epsilon \in \{0.1, 0.01, 0.001\}$

→ The qAER reaches a plateau at SNR = 10dB for both schemes

Time of execution

We generated some pairs $(\text{qAER}(\epsilon), T_{\text{mean}}(\epsilon))$ with different values of ϵ

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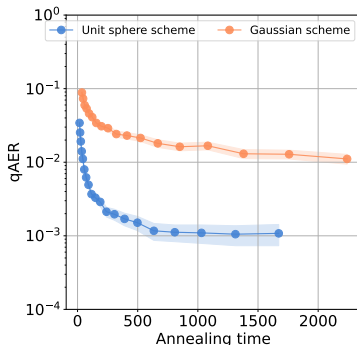


Figure 5: qAER against the annealing time at SNR = 15dB

→ The unit sphere scheme reaches faster a 10^{-3} error-rate than the Gaussian scheme.

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Contributions

- We designed a quantum annealing algorithm for covariance based AUD
- We introduced the qAER metric
- The choice of the matrix $\mathbf{P}$ appears to be an important decision in terms of computational time required by QA.

Next steps

- Propose preamble matrices $\mathbf{P}$ that meet stricter constraints (compressed sensing framework)

Thank you! Questions ?